

MATH 345 Skeleton Notes

Unit 4: Orthogonality and least squares

Section 4.2: Orthogonal projection

© 2024–2026 Jacob Denson, Kaitlyn Phillipson, Sebastien Roch. Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Orthogonal complements and projections

Definition: The orthogonal complement of a subspace

If U is a subspace of $\mathbb{R}^n$, the orthogonal complement of U is the subset of $\mathbb{R}^n$ given by

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \text{ for all } \mathbf{y} \text{ in } U\}$$

Theorem

The following properties of the The orthogonal complement of a subspace hold:

1. For any subspace U of $\mathbb{R}^n$, $U^\perp$ is a subspace of $\mathbb{R}^n$
2. $\{0\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{0\}$
3. If $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, then

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{x}_i = 0 \text{ for } i = 1, 2, \dots, k\}$$

Activity: Finding an orthogonal complement (U4-LO2)

Find $U^\perp$ if $U = \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Lemma: Expanding an orthogonal set

Let $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ be an the definition in $\mathbb{R}^n$. Given $\mathbf{x}$ in $\mathbb{R}^n$, write

$$\mathbf{f}_{m+1} = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m$$

Then:

- $\mathbf{f}_{m+1} \cdot \mathbf{f}_k = 0$ for $k = 1, 2, \dots, m$
- If $\mathbf{x} \notin \text{span}\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then $\mathbf{f}_{m+1} \neq \mathbf{0}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_m, \mathbf{f}_{m+1}\}$ is an orthogonal set.

Why is this true?

For convenience, let $t_i = \frac{\mathbf{x} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$ for each i . Given $1 \leq k \leq m$, we calculate that

$$\begin{aligned} \mathbf{f}_{m+1} \cdot \mathbf{f}_k &= (\mathbf{x} - t_1 \mathbf{f}_1 - t_2 \mathbf{f}_2 - \dots - t_m \mathbf{f}_m) \cdot \mathbf{f}_k \\ &= \mathbf{x} \cdot \mathbf{f}_k - t_1 (\mathbf{f}_1 \cdot \mathbf{f}_k) - \dots - t_k (\mathbf{f}_k \cdot \mathbf{f}_k) - \dots - t_m (\mathbf{f}_m \cdot \mathbf{f}_k) \end{aligned}$$

Since $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is an orthogonal set, $\mathbf{f}_i \cdot \mathbf{f}_k = 0$ for all $i \neq k$. So

$$\begin{aligned} \mathbf{f}_{m+1} \cdot \mathbf{f}_k &= \mathbf{x} \cdot \mathbf{f}_k - t_k (\mathbf{f}_k \cdot \mathbf{f}_k) \\ &= \mathbf{x} \cdot \mathbf{f}_k - \frac{\mathbf{x} \cdot \mathbf{f}_k}{\|\mathbf{f}_k\|^2} \cdot \|\mathbf{f}_k\|^2 \\ &= \mathbf{x} \cdot \mathbf{f}_k - \mathbf{x} \cdot \mathbf{f}_k = 0 \end{aligned}$$

This proves the first property of the lemma. To prove the second, we note that if $\mathbf{f}_{m+1}$ was zero, then $\mathbf{x}$ *would* be in the span of the vectors $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$. So if $\mathbf{x}$ is not in the span, then $\mathbf{f}_{m+1}$ can not be equal to zero. And we have proved $\mathbf{f}_{m+1}$ is orthogonal to the vectors $\mathbf{f}_1, \dots, \mathbf{f}_m$, so that $\{\mathbf{f}_1, \dots, \mathbf{f}_{m+1}\}$ is an orthogonal set.

Activity: Extending an orthogonal set to a basis (U4-L02)

Use Expanding an orthogonal set to extend the orthogonal set $\{\mathbf{f}_1, \mathbf{f}_2\}$ in $\mathbb{R}^3$ to an orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$, where

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

using the vector $\mathbf{x} = (1, 1, 2)$ as in that lemma.

Definition: The orthogonal projection onto a subspace

Let U be a subspace of $\mathbb{R}^n$. Then

the **orthogonal projection** of the vector $\mathbf{x} \in \mathbb{R}^n$ onto U is the vector $\mathbf{y} = \text{proj}_U(\mathbf{x}) \in U$ which is *closest* to $\mathbf{x}$, in the sense that for any $\mathbf{z} \in U$ with $\mathbf{z} \neq \mathbf{y}$, $\|\mathbf{x} - \mathbf{y}\| < \|\mathbf{x} - \mathbf{z}\|$. The **residual** is the difference between the original vector and its projection:

$$\mathbf{r} = \mathbf{x} - \mathbf{y} = \mathbf{x} - \text{proj}_U(\mathbf{x}).$$

Orthogonal projection decomposes the original vector into a vector in the subspace and a residual perpendicular to the subspace.

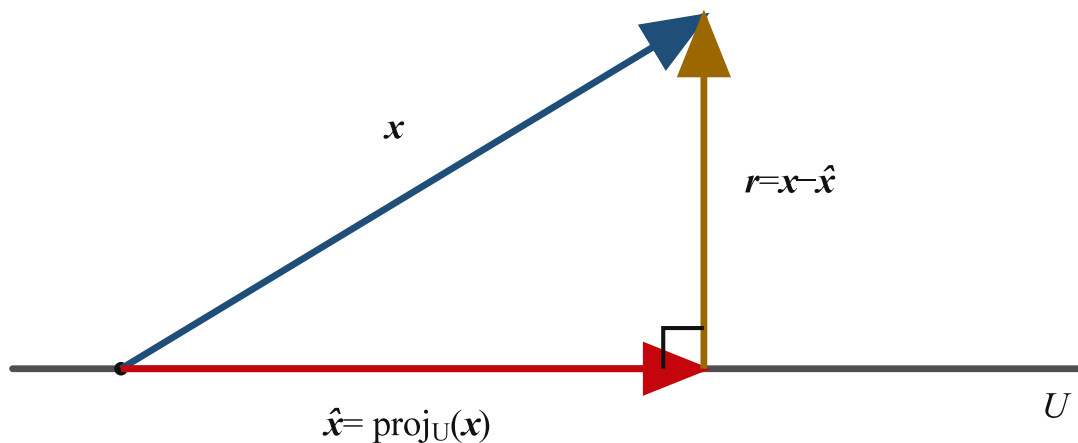


Figure. Projection and residual. Orthogonal projection writes $\mathbf{x} = \hat{\mathbf{x}} + \mathbf{r}$, where $\hat{\mathbf{x}} \in U$ and $\mathbf{r} \in U^\perp$.

The following theorem gives us a practical way to calculate orthogonal projections.

Theorem: Projection theorem

Let U be a subspace of $\mathbb{R}^n$ with orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$. For any vector $\mathbf{x} \in \mathbb{R}^n$, define

$$\mathbf{y} = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m.$$

Then $\mathbf{y} = \text{proj}_U(\mathbf{x})$. The residual

$$\mathbf{r} = \mathbf{x} - \mathbf{y} = \mathbf{x} - \text{proj}_U(\mathbf{x})$$

lies in $U^\perp$. In particular, if $\mathbf{x}$ is already in U , then $\mathbf{x}$ equals the displayed sum.

Why is this true?

If $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ is an orthogonal basis for U , and we define

$$\mathbf{y} = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m,$$

then $\mathbf{y}$ is evidently a linear combination of the basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, so $\mathbf{y} \in U$. And Expanding an orthogonal set states that $\mathbf{x} - \mathbf{y} \in U^\perp$. Thus if $\mathbf{z} \in U$ and $\mathbf{z} \neq \mathbf{y}$, the Pythagorean theorem (Pythagoras' theorem) tells us that

$$\begin{aligned} \|\mathbf{x} - \mathbf{z}\|^2 &= \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\|^2 \\ &= \|\mathbf{x} - \mathbf{y}\|^2 + \|\mathbf{y} - \mathbf{z}\|^2. \end{aligned}$$

Since $\mathbf{y} \neq \mathbf{z}$, we find that

$$\|\mathbf{x} - \mathbf{z}\|^2 > \|\mathbf{x} - \mathbf{y}\|^2,$$

and taking square roots gives that

$$\|\mathbf{x} - \mathbf{z}\| > \|\mathbf{x} - \mathbf{y}\|.$$

Thus the vector $\mathbf{y}$ we have defined is equal to $\text{proj}_U(\mathbf{x})$. Since we have already verified that $\mathbf{x} - \mathbf{y} = \mathbf{x} - \text{proj}_U(\mathbf{x})$ is in $U^\perp$, the theorem follows.

Activity: Projecting onto a span (U4-LO3)

Find $\text{proj}_U \mathbf{x}$ where $U = \text{span}\{(1, 1, 0), (1, -1, 0)\}$ and $\mathbf{x} = (2, 0, 1)$ in $\mathbb{R}^3$.

Activity: Projection onto a line (U4-LO3)

Let

$$\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad L = \text{span}\{\mathbf{u}\}.$$

1. Compute $\text{proj}_L(\mathbf{x})$.
2. Compute the residual $\mathbf{r} = \mathbf{x} - \text{proj}_L(\mathbf{x})$.
3. Check that $\mathbf{r} \cdot \mathbf{u} = 0$.

Activity: Projection onto a line in code (U4-LO3, U4-LO7)

The following code computes the same projection and residual as the previous activity.

```
import numpy as np

u = np.array([3.0, 4.0])
x = np.array([2.0, 5.0])

y = (x @ u) / (u @ u) * u
r = x - y
```

Activity: Projection onto a line in code (U4-LO3, U4-LO7) (continued)

```
y, r, r @ u
```

1. Which variable stores the projection?
2. Which variable stores the residual?
3. Why should $r @ u$ be close to zero?
4. What does $(x @ u) / (u @ u) * u$ compute?

Let $\mathbf{v}$ be a unit vector. By Projection theorem,

$$\text{proj}_{\text{span}\{\mathbf{v}\}}(\nabla f(\mathbf{a})) = (\nabla f(\mathbf{a}) \cdot \mathbf{v})\mathbf{v}.$$

By the theorem,

$$\text{proj}_{\text{span}\{\mathbf{v}\}}(\nabla f(\mathbf{a})) = D_{\mathbf{v}}f(\mathbf{a})\mathbf{v}.$$

Thus $D_{\mathbf{v}}f(\mathbf{a})$ is the signed projection coefficient of the gradient in the oriented direction $\mathbf{v}$. Its absolute value is the length of the projection. The Cauchy–Schwarz inequality in The Cauchy-Schwarz inequality gives

$$|D_{\mathbf{v}}f(\mathbf{a})| \leq \|\nabla f(\mathbf{a})\|.$$

Activity: Directional derivative as a projection coefficient (U4-LO3, U3-LO8)

Reopen Alignment with the gradient (U3-LO8, U1-LO1). For each listed unit vector $\mathbf{v}_i$, compute

$$\text{proj}_{\text{span}\{\mathbf{v}_i\}}(\nabla f(\mathbf{a})) = D_{\mathbf{v}_i}f(\mathbf{a})\mathbf{v}_i.$$

1. Which projection coefficients are positive, and which are negative?
2. Why do $\mathbf{v}_3$ and $\mathbf{v}_4$ give the same projection vector but opposite directional derivatives?
3. Reopen fig-u3-gradient-directions. How does the figure distinguish greatest increase, greatest decrease, and zero change?

Orthogonal projection can also be described by minimizing a distance function.

Example

A common optimization problem that we have already encountered in The orthogonal projection onto a subspace involves taking a subspace $U \subseteq \mathbb{R}^n$ and a given vector $\mathbf{x}_0 \in \mathbb{R}^n$, and finding the vector $\mathbf{x}^* \in U$ that is *closest* to $\mathbf{x}_0$, i.e., finding the vector that minimizes the scalar-valued function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ on U , defined by setting $g(\mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|$. Then $\mathbf{x}^*$ is the orthogonal projection of $\mathbf{x}_0$ onto U , and we have seen in Projection theorem that if U has an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then

$$\mathbf{x}^* = \frac{\mathbf{x}_0 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 + \frac{\mathbf{x}_0 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 + \dots + \frac{\mathbf{x}_0 \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2} \mathbf{f}_m.$$

In Least squares: three viewpoints, we return to this viewpoint and derive the normal equations by setting a gradient equal to zero.

Activity: Projection as optimization (U4-LO3, U3-LO9)

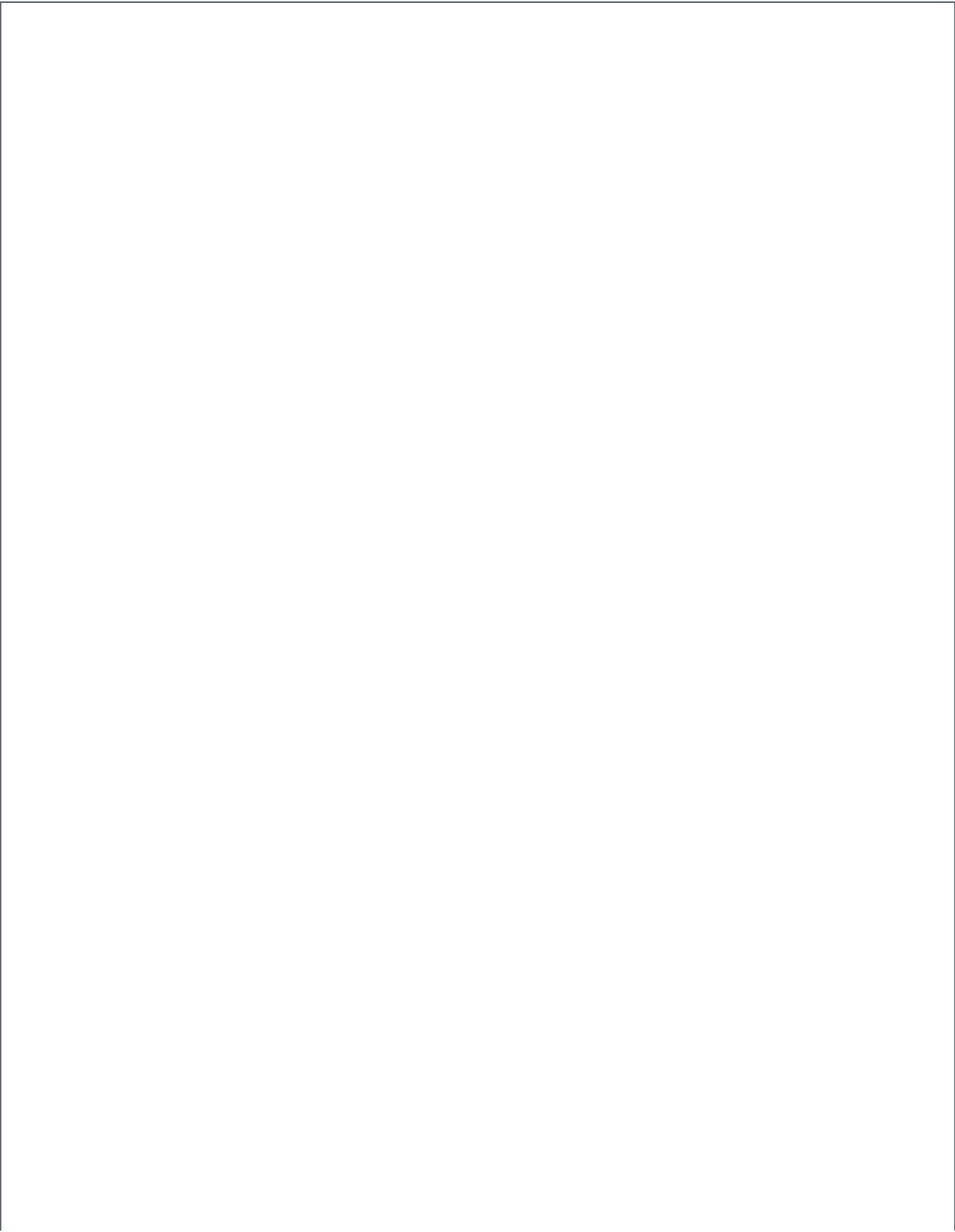
Let

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^2, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A general point of W has the form

$$\mathbf{w}(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

1. Write the squared distance $g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2$.
2. Compute $g'(t)$ and find the critical point of g .
3. Use $g''(t)$ to classify the critical point.
4. What point of W is closest to $\mathbf{b}$?
5. How does this agree with the figure and Projection theorem?



Projection and attention both use dot products to score directions and then use the resulting numbers in a linear combination.

Let

$$Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_k]$$

have orthonormal columns. Then

$$Q^T \mathbf{x} = \begin{bmatrix} \mathbf{q}_1^T \mathbf{x} \\ \vdots \\ \mathbf{q}_k^T \mathbf{x} \end{bmatrix}$$

collects the directional coefficients of $\mathbf{x}$. Because each $\mathbf{q}_i$ has length 1, these dot products are the coordinates used in the projection. If $\mathbf{x}$ also has length 1, they are cosine similarities with the directions $\mathbf{q}_i$.

Multiplying by Q combines the directions using those coefficients:

$$Q(Q^T \mathbf{x}) = \sum_{i=1}^k (\mathbf{q}_i^T \mathbf{x}) \mathbf{q}_i = \text{proj}_{\text{col}(Q)}(\mathbf{x}).$$

Thus projection follows the pattern

score with $Q^T \longrightarrow$ combine with Q .

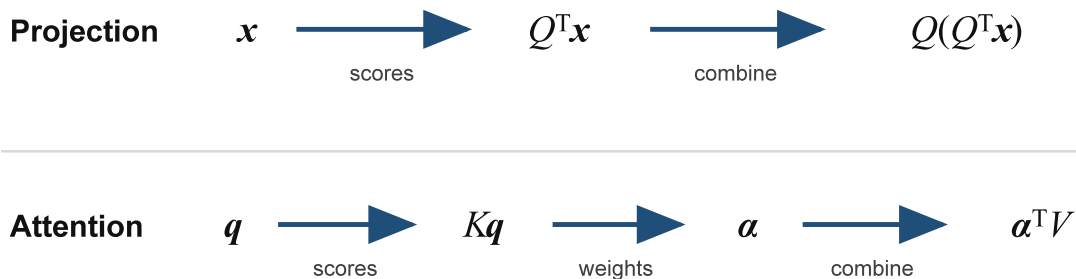


Figure. Score, then combine. Projection uses $Q^T \mathbf{x}$ to score orthonormal directions and $Q(Q^T \mathbf{x})$ to combine them. Attention uses query-key scores, converts them into weights, and combines value vectors.

Attention has the same broad pattern. Query-key dot products produce scores, a weighting rule converts the scores into weights, and the weights combine value vectors. Projection scores and combines the same orthonormal directions. Attention scores key vectors and then combines value vectors, which may be different vectors. Projection also chooses its coefficients so that the residual is orthogonal to the subspace. Attention is therefore not generally an orthogonal projection, but both computations are built from dot products and linear combinations.

Activity: Dot-product scores as projection coordinates (U4-LO1, U4-LO3)

Let

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix},$$

and let

Activity: Dot-product scores as projection coordinates (U4-LO1, U4-LO3)
(continued)

$$Q = [\mathbf{q}_1 \ \mathbf{q}_2] .$$

1. Verify that the columns of Q are orthonormal.
2. Compute the score vector

$$\mathbf{c} = Q^T \mathbf{x} .$$

3. Compute

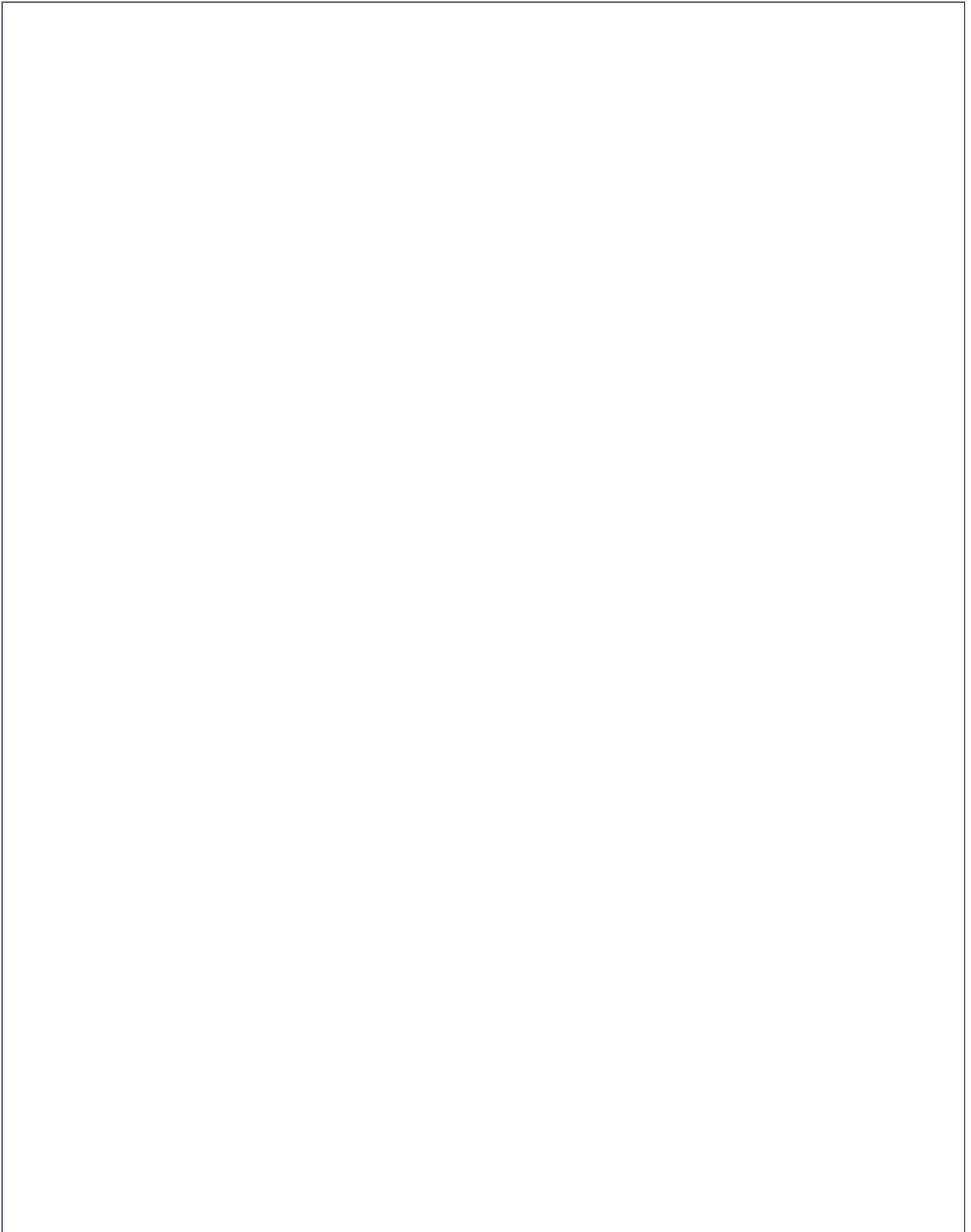
$$\hat{\mathbf{x}} = Q\mathbf{c} .$$

4. Compute the residual

$$\mathbf{r} = \mathbf{x} - \hat{\mathbf{x}} ,$$

and verify that $Q^T \mathbf{r} = \mathbf{0}$.

5. Explain why the entries of $\mathbf{c}$ can be read both as dot-product scores and as coordinates of the projection.
6. Explain how the calculation follows the pattern “score, then combine.” What extra condition identifies $\hat{\mathbf{x}}$ as an orthogonal projection?



Gram-Schmidt orthogonalization

Gram-Schmidt is a sequence of projection, subtraction, measurement, and normalization steps. Use *Perfectly Normal* to practice the same sequence geometrically. The game uses an original column $\mathbf{a}_j$, its projection $\mathbf{p}_j$, the residual $\mathbf{v}_j$, the new unit direction $\mathbf{q}_j$, and coefficients r_{ij} .

Algorithm: Gram-Schmidt algorithm

Input:	A matrix
---------------	----------

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$$

with linearly independent columns.

For $j = 1, \dots, n$:

1. **Current column.** Start with $\mathbf{a}_j$.

2. **Project.** For each $i < j$, compute

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j.$$

Then assemble

$$\mathbf{p}_j = \sum_{i < j} r_{ij} \mathbf{q}_i.$$

For the first column, $\mathbf{p}_1 = \mathbf{0}$.

3. **Subtract.**

$$\mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j.$$

4. **Measure.**

$$r_{jj} = \|\mathbf{v}_j\|.$$

5. **Normalize.**

$$\mathbf{q}_j = \frac{\mathbf{v}_j}{r_{jj}}.$$

Output:

The vectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal, and

$$\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$$

at every stage.

Warning

If $r_{jj} = 0$, then $\mathbf{a}_j$ is already in the span of the earlier columns. The input columns are not linearly independent, and no new unit direction is produced.

The same algorithm in orthogonal form.

Repeatedly applying

Expanding an orthogonal set gives the theorem below. The theorem first constructs orthogonal residuals $\mathbf{f}_j$, while the algorithm above normalizes each residual to obtain $\mathbf{q}_j$. The two forms agree because

$$\mathbf{q}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|}$$

and

$$\frac{\mathbf{a}_j^T \mathbf{f}_i}{\|\mathbf{f}_i\|^2} \mathbf{f}_i = (\mathbf{q}_i^T \mathbf{a}_j) \mathbf{q}_i.$$

Theorem: The Gram-Schmidt orthogonalization algorithm

Let U be a subspace of $\mathbb{R}^n$. Then U has an orthogonal basis.

Given a Bases of Vector Spaces $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ of U , we can iteratively define an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ of U by defining

$$\begin{aligned}\mathbf{f}_1 &= \mathbf{x}_1 \\ \mathbf{f}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 \\ \mathbf{f}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2\end{aligned}$$

and so on, i.e., for $2 \leq k \leq m$ defining

$$\mathbf{f}_k = \mathbf{x}_k - \frac{\mathbf{x}_k \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{x}_k \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{x}_k \cdot \mathbf{f}_{k-1}}{\|\mathbf{f}_{k-1}\|^2} \mathbf{f}_{k-1}.$$

Then

- $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\}$ is an orthogonal basis of U .
- $\text{span}\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\} = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ for each $k = 1, 2, \dots, m$.

Activity: Gram-Schmidt for a row space (U4-LO2)

Apply the Gram-Schmidt algorithm to the columns of A^T , where

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 3 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

These columns are the rows of A , written as column vectors. At each stage, record the projection coefficients, projection, residual, residual length, and normalized direction. Give an orthogonal basis and an orthonormal basis for $\text{row}(A)$ (recall Column and Row Space).

1. Compute $\mathbf{v}_1$, r_{11} , and $\mathbf{q}_1$.
2. Compute r_{12} , $\mathbf{p}_2$, $\mathbf{v}_2$, r_{22} , and $\mathbf{q}_2$.
3. Compute r_{13} , r_{23} , $\mathbf{p}_3$, $\mathbf{v}_3$, r_{33} , and $\mathbf{q}_3$.
4. Give an orthogonal basis from the residuals.

Activity: Gram–Schmidt for a row space (U4-LO2) (continued)

5. Give an orthonormal basis from the unit vectors.
6. Explain why both bases span $\text{row}(A)$.

Activity: The closest point on a plane (U4-LO3)

Find the point in the plane defined by the equation $2x + y - z = 0$, which is closest to the point $(1, 2, -1)$.

